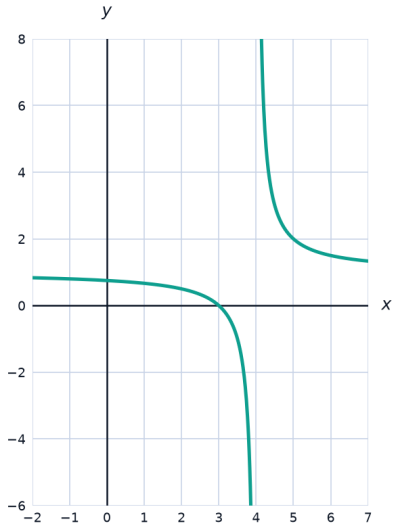
Rational Functions & Asymptotes

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Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Vertical asymptotes and holes**

Find the vertical asymptotes and holes of $f(x) = \frac{x^2 - 9}{x^2 - x - 12}$.



2 **[Foundational]** • **Target: Vertical asymptotes and holes**

Find the y-coordinate of the hole in the function from the previous problem.

3 **[Foundational]** • **Target: Horizontal and slant asymptotes**

Find the horizontal asymptote of $f(x) = \frac{3x^2 - 2x + 1}{5x^2 + 4}$, using the degrees of numerator and denominator.

4 **[Foundational]** • **Target: Horizontal and slant asymptotes**

Find the horizontal asymptote of $g(x) = \frac{2x + 7}{x^2 - 1}$.

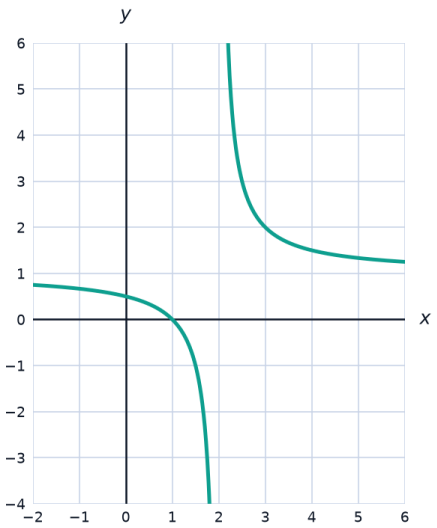
5 **[Foundational]** • **Target: Horizontal and slant asymptotes**

Explain why $h(x) = \frac{x^2 + 1}{x - 2}$ has a slant (oblique) asymptote instead of a horizontal one, and find it using polynomial division.

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6 [Foundational] • Target: Visualizing a rational function's asymptotes

The graph shows $f(x) = \frac{1}{x-2} + 1$, a transformed reciprocal function.

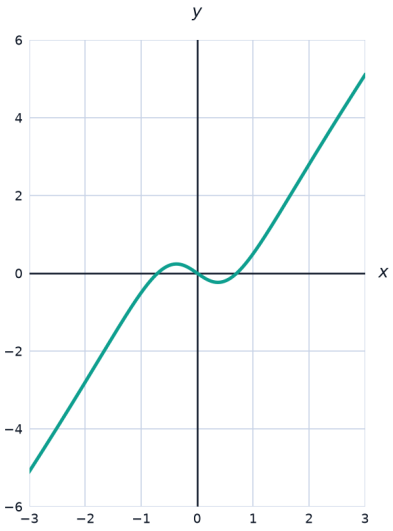


a Identify the vertical asymptote shown.

b Identify the horizontal asymptote shown.

7 [Foundational] • Target: End behavior of rational functions

Describe the end behavior of $f(x) = \frac{2x^3 - x}{x^2 + 1}$ as $x \to \pm\infty$ (note the numerator's degree exceeds the denominator's by 1, so the function behaves like its slant asymptote).

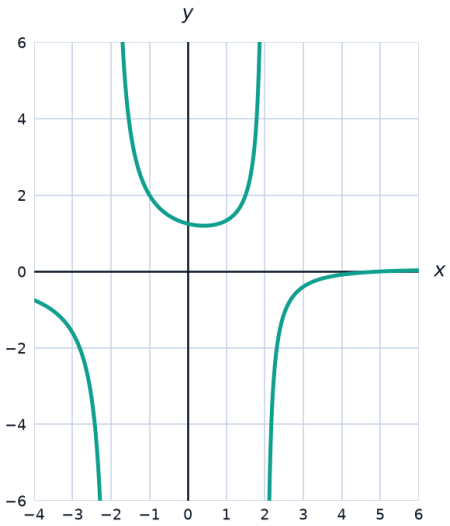


Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 **[Intermediate]** • **Target: Domain and intercepts**

Find the domain, x-intercept(s), and y-intercept of $f(x) = \frac{x-5}{x^2-4}$.



9 **[Intermediate]** • **Target: Solving rational equations and inequalities**

Solve $\frac{x+1}{x-3} = 2$, checking that the solution doesn't create a zero denominator.

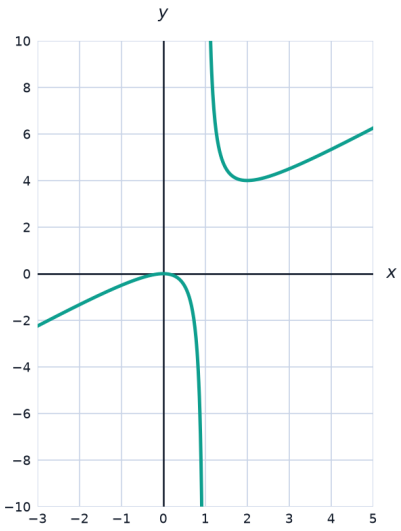
10 **[Intermediate]** • **Target: Solving rational equations and inequalities**

Solve the inequality $\frac{x-1}{x+2} > 0$ by determining the sign of the expression on each interval created by the zero and the excluded value.

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11 **[Intermediate]** • **Target: Visualizing a function with a slant asymptote**

The graph shows $f(x) = \frac{x^2}{x-1}$, which has a slant asymptote.



a Find the vertical asymptote and confirm it matches the graph.

b Use polynomial division to find the slant asymptote equation.

c Explain why this function has no horizontal asymptote.

12 **[Intermediate]** • **Target: Building a rational function from given features**

Construct a rational function with a vertical asymptote at $x = 3$, a hole at $x = -1$, and a horizontal asymptote at $y = 2$.

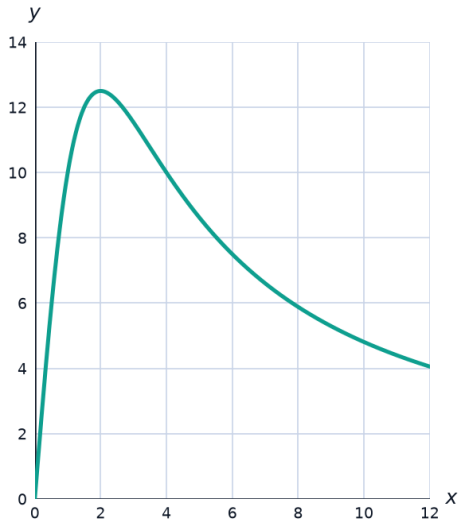
13 **[Intermediate]** • **Target: Building a rational function from given features**

Explain why a rational function cannot have both a horizontal asymptote and a slant asymptote at the same time.

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14 **[Intermediate]** • **Target: Rational functions in context**

The concentration of a drug in the bloodstream t hours after a dose is modeled by $C(t) = \frac{50t}{t^2 + 4}$ (mg/L). Find the horizontal asymptote and interpret it in context.



Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Solving rational equations and inequalities** • **Cross-Lesson: 4.01 Inverse & Composite Functions**

Level 3 · AdvancedLet $R(x) = \frac{2x^2 - 8}{x^2 - 5x + 6}$. (a) Find the domain of $R(x)$. (b) Identify any holes and state their coordinates. (c) Write the equations of all vertical and horizontal asymptotes.

16 **[Challenging]** • **Target: Visualizing a function with a slant asymptote** • **Cross-Lesson: 4.02 Piecewise-Defined & Implicit Functions**

Find all vertical, horizontal, and slant asymptotes and any removable discontinuities (holes) of the rational function $f(x) = \frac{3x^2 + 1}{x^2 - 4}$. Describe its end behavior using limits.

17 **[Challenging]** • **Target: Building a rational function from given features** • **Cross-Lesson: 4.03 Parametric Equations & Modeling**

Find all vertical and horizontal asymptotes, the exact coordinates of any holes, and all intercepts of the rational function $R(x) = \frac{2x^2 + 5x - 3}{x^2 + 4x + 3}$.

18 **[Challenging]** • **Target: Rational functions in context** • **Cross-Lesson: 1.01 Polynomial End Behavior & Zeros**

Level 2 · MediumThe function $f(x) = x^3 - 3x^2 - 9x + 2$ has a local maximum and a local minimum. Determine the intervals on which f is concave up and concave down, and find the coordinates of the inflection point.

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19 [Challenging] • Target: Vertical asymptotes and holes • Cross-Lesson: 1.03 Rates of Change & Concavity

Level 1 · FoundationsConsider the restricted quadratic function $g(x) = 2(x-3)^2 + 5$ defined on the domain $[3, \infty)$. Explain why g is invertible on this domain, find the formula for $g^{-1}(x)$, and state the domain of g^{-1}.

20 [Challenging] • Target: Horizontal and slant asymptotes • Cross-Lesson: 2.01 Exponential Functions & Growth Models

Level 3 · AdvancedExperimental laboratory data pairs (x, y) are transformed by computing $Y = \ln(y)$. A linear regression performed on (x, Y) yields the best-fit line $Y = -0.42x + 2.85$. Write the corresponding non-linear model for y as an exponential function of x in the form $y = ab^x$, giving a and b to three decimal places. Then predict the value of y when $x = 5$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Vertical asymptotes and holes

Find the hole and vertical asymptote of $f(x) = \frac{(x+2)(x-5)}{(x+2)(x-6)}$.

a. Hole at $x = 6$; vertical asymptote at $x = -2$.

b. Hole at $x = -2$; vertical asymptote at $x = 6$.

c. Vertical asymptotes at $x = -2$ and $x = 6$, no hole.

d. Hole at $x = 5$; vertical asymptote at $x = 6$.

22 [Foundational] • Target: Horizontal and slant asymptotes

Find the slant asymptote of $h(x) = \frac{x^2 + x + 7}{x - 7}$.

a. $y = x + 9$

b. $y = x + 7$

c. $y = x - 8$

d. $y = x + 8$

23 [Foundational] • Target: Visualizing a rational function's asymptotes

Solve the inequality $\frac{x+6}{x-3} > 0$.

a. $(-\infty, 3) \cup (-6, \infty)$

b. $(-\infty, -6) \cup (3, \infty)$

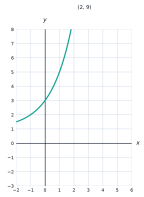
c. $[-6, 3]$

d. $(-6, 3)$

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24 [Foundational] • Target: End behavior of rational functions

The graph of the function $f(x) = a \cdot 2^{x-h} + k$ is shown below. The graph passes through the points $(2, 9)$ and $(4, 33)$. The horizontal asymptote is $y = 1$. Find the value of a.



a. 4

b. 2

c. 1/2

d. 8

25 [Foundational] • Target: Domain and intercepts

Consider the function $f(x) = \frac{x^3 - 8}{x^2 - 4}$. We are interested in the behavior of this function near its discontinuities. Let $x = a$ be a value where the function has a removable discontinuity, and let $x = b$ be a value where the function has a non-removable discontinuity (vertical asymptote). What is the value of the limit of $f(x)$ as x approaches a, and what is the value of $\lim_{x \to b^-} f(x)$?

a. Limit at a is 3, $\lim_{x \to b^-} f(x) = -\infty$

b. Limit at a is 3, $\lim_{x \to b^-} f(x) = \infty$

c. Limit at a is 1, $\lim_{x \to b^-} f(x) = -\infty$

d. Limit at a is 1, $\lim_{x \to b^-} f(x) = \infty$

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26 [Foundational] • Target: Solving rational equations and inequalities

Consider the rational function $f(x) = \frac{ax^2 - 4x + 1}{x^2 - bx + 2}$. The graph of $y = f(x)$ has a vertical asymptote at $x = 1$ and a horizontal asymptote at $y = 2$. If $f(0) = \frac{1}{2}$, find the value of a.

a. 1

b. 2

c. 3

d. 4

27 [Foundational] • Target: Visualizing a function with a slant asymptote

Find all rational functions with vertical asymptotes at $x = \pm 1$ and horizontal asymptote $y = 0$.

a. $\frac{x}{x^2 - 1}$ only

b. $\frac{1}{x - 1}$ only

c. Any polynomial

d. Functions of the form $\frac{k}{x^2 - 1}$

28 [Foundational] • Target: Building a rational function from given features

The graph of the rational function $g(x) = \frac{ax^2 + bx + 1}{x^2 - 4}$ has a horizontal asymptote at $y = 2$ and a vertical asymptote at $x = 2$. Given that $g(0) = -1/4$, find the value of a.

a. 1

b. 2

c. 3

d. 4

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Rational functions in context

The function f is defined by $f(x) = \frac{ax + 8}{cx + d}$ for all real numbers $xeq -\frac{d}{c}$, where a, c, and d are nonzero constants. In the standard (x, y) coordinate plane, the graph of $y = f(x)$ has a horizontal asymptote at $y = 2$, a vertical asymptote at $x = 3$, and an x-intercept at $(-4, 0)$. What is the value of $f^{-1}(4)$?

a. 10

b. 16

c. -2

d. 2

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30 **[Intermediate]** • **Target: Vertical asymptotes and holes**

Consider the rational function $f(x) = \frac{x^2 - ax + 6}{x - b}$. If the graph of $f(x)$ has a removable discontinuity at $x = 2$ and a vertical asymptote at $x = 3$, determine the values of a and b.

a. a = 5, b = 3

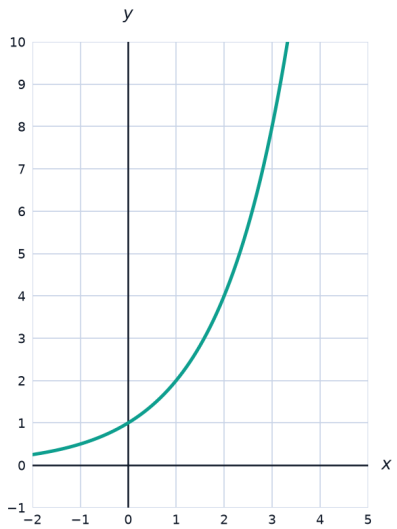
b. a = -5, b = 3

c. a = 5, b = 2

d. a = -5, b = 2

31 **[Intermediate]** • **Target: Horizontal and slant asymptotes**

Consider the function $f(x) = 2^x$. Which of the following points lies on the graph of $f(x)$?



a. $(1, 1)$

b. $(2, 4)$

c. $(3, 6)$

d. $(0, 2)$

32 **[Intermediate]** • **Target: Visualizing a rational function's asymptotes**

Find the x-intercepts and vertical asymptotes of $f(x) = \frac{x^2 - x - 6}{x^2 - 4}$.

a. x-int: $x = 3, -2$ (but $x = -2$ is a hole); VA: $x = 2$

b. x-int: $x = 3$; VA: $x = \pm 2$

c. x-int: $x = 3, -2$; VA: $x = \pm 2$

d. x-int: $x = 3$; VA: $x = 2$ only

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33 **[Intermediate]** • **Target: End behavior of rational functions**

Which function has a slant asymptote?

a. $f(x) = \frac{1}{x^2}$

b. $f(x) = \frac{x}{x^2 + 1}$

c. $f(x) = \frac{x^2 + 1}{x}$

d. $f(x) = \frac{x + 1}{x + 2}$

34 **[Intermediate]** • **Target: Domain and intercepts**

Which function has no vertical asymptote?

a. $f(x) = \frac{x}{x - 1}$

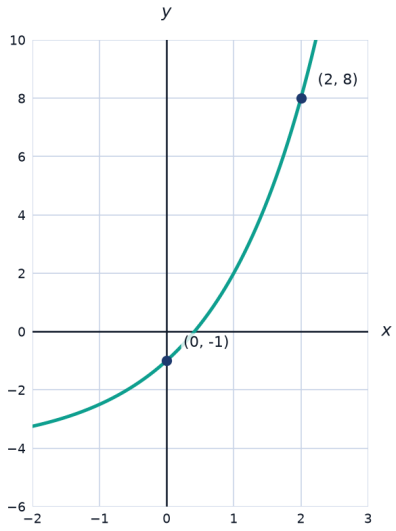
b. $f(x) = \frac{1}{x^2}$

c. $f(x) = \frac{x^2 + 1}{x^2 + 1} = 1$

d. $f(x) = \frac{1}{x}$

35 **[Intermediate]** • **Target: Solving rational equations and inequalities**

The graph of the exponential function $f(x) = a \cdot b^x + c$ is shown in the xy-plane, where a, b, and c are constants with $b > 0$ and $beq1$. The function has a horizontal asymptote at $y = -4$ and passes through the points $(0, -1)$ and $(2, 8)$. What is the value of $f(-1)$?



a. $-\frac{5}{2}$

b. $-\frac{7}{2}$

c. $-\frac{13}{4}$

d. $\frac{3}{2}$

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36 [Intermediate] • Target: Visualizing a function with a slant asymptote

Consider the function $f(x) = \log_3(ax + b)$, where a and b are positive constants. If the graph of $y = f(x)$ passes through the points $(1, 2)$ and $(10, 3)$, find the value of a.

a. 3

b. 1

c. 1/3

d. 9

37 [Intermediate] • Target: Building a rational function from given features

Consider the exponential function $f(x) = a \cdot b^x$ where a and b are positive constants. The graph of $y = f(x)$ passes through the points $P = (1, 6)$ and $Q = (3, 54)$. A related function $g(x)$ is defined such that $g(x) = \log_b\left(\frac{f(x)}{a}\right)$. If the point R on the graph of $y = g(x)$ has an x-coordinate of 2, what is the y-coordinate of R?

a. 2

b. 3

c. $\log_3 9$

d. $\frac{1}{2}\log_3 18$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Rational functions in context • Cross-Lesson: 1.01 Polynomial End Behavior & Zeros

The graph of the function $g(x) = a \cdot b^x$ passes through the points $(1, 6)$ and $(3, 24)$. If a and b are positive constants, what is the value of a?

a. 1.5

b. 2

c. 3

d. 6

39 [Challenging] • Target: Vertical asymptotes and holes • Cross-Lesson: 1.03 Rates of Change & Concavity

A scientist is modeling the population growth of a rare species of bioluminescent fungi. The population, $P(t)$, in thousands, after t months is given by the function $P(t) = P_0e^{kt}$, where P_0 is the initial population and k is the growth constant. Observations show that after 2 months, the population has doubled from its initial size. Furthermore, after 6 months, the population is 16,000. What is the initial population P_0?

a. 2,000

b. 4,000

c. 8,000

d. 1,000

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40 [Challenging] • Target: Horizontal and slant asymptotes • Cross-Lesson: 2.01 Exponential Functions & Growth Models

The graph of the exponential function $f(x) = a \cdot b^x$ passes through the points $(1, 12)$ and $(2, 36)$. If $b > 0$ and $beq1$, find the value of a.

a. 4

b. 3

c. 12

d. 1

41 [Challenging] • Target: Visualizing a rational function's asymptotes • Cross-Lesson: 2.02 Logarithmic Functions & Properties

The graph of the exponential function $f(x) = a \cdot b^x$ passes through the points $(1, 6)$ and $(2, 18)$. What is the value of b?

a. 2

b. 3

c. 4

d. 6

42 [Challenging] • Target: End behavior of rational functions • Cross-Lesson: 3.01 Trigonometric Functions & the Unit Circle

The graph of the exponential function $f(x) = a \cdot 2^x$ passes through the point $(3, 40)$. What is the value of a?

a. $\frac{5}{2}$

b. 5

c. 10

d. 20

43 [Challenging] • Target: Domain and intercepts • Cross-Lesson: 3.02 Polar Coordinates & Polar Graphs

The graph of the function $f(x) = A\sin(Bx + C) + D$ is shown below. The maximum value of the function is 5 and the minimum value is -1. The period of the function is $\frac{\pi}{2}$, and the graph passes through the point $(0, 2)$. Find the value of A.

a. 2

b. 3

c. 4

d. 6

44 [Challenging] • Target: Solving rational equations and inequalities • Cross-Lesson: 4.01 Inverse & Composite Functions

Consider the exponential function $f(x) = 4 \cdot 2^x$. What is the value of $f(3)$?

a. 12

b. 16

c. 32

d. 24

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45 [Challenging] • Target: Visualizing a function with a slant asymptote • Cross-Lesson: 4.02 Piecewise-Defined & Implicit Functions

Which function is decreasing on all of $\mathbb{R}$?

a. $y = e^{x}$

b. $y = x^{2}$

c. $y = e^{-x}$

d. $y = |x|$

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